

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Java and Spring Boot microservices**, Hibernate/JPA, event-driven architectures, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Java, SQL, Python, C, C++.
- Backend & APIs** – Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST, gRPC, Spring Batch, Design Patterns, Service-Oriented Architecture (SOA).
- AI & Agents** – Python (FastAPI, LangGraph, LangChain), LLM agents (OpenAI/Gemini/Claude), MCP, RAG, Milvus, pgvector.
- Streaming & Data** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, Batch Processing.
- Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Amazon S3, Caching.
- Cloud & Infrastructure** – GCP, AWS, Docker, Kubernetes, CI/CD, Maven, Bazel, ELK, Grafana, New Relic, Datadog, PostHog, LangSmith.
- Core Engineering** – Distributed Systems, HLD/LLD, Multithreading, Concurrency, Sharding, Mentorship, JUnit, Mockito.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building AssortSmart so retailers decide **what to buy, how much, and for which stores** with a planner chat (**Python, LangGraph, MCP**) and shared **Spring Boot** write APIs, targeting plan turnaround from days toward **under 1 hour** and search from **1 to ≥ 20 configs/plan**.
- Cutting measured clustering-run failures from **8.5% (37/437 on kik)** toward **under 2%** (target) with **14** audited read-only tools and **3** human confirm steps so the agent never writes SQL.
- Replacing a projected **12B-row** store-week design with a **$\sim 25\text{M}$ -row** aggregate so month rollups stay **sub-second** and cell edits stay **$\sim 0.4\text{ ms}$** (measured) without exploding the table.
- Adopting per-tenant **ClickHouse (63 tables / 8 layers)**, insert-only, refresh by swapping partitions, agent read-only) as the planning store after a same-rows POC cut **250M-row** pivots from **189s to 12.3s** ($\sim 15.5\text{x}$, measured).
- Tech** – Java, Spring Boot, Python, FastAPI, LangGraph, MCP, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, Docker.

Software Development Engineer 2

UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned design and architecture for Uber's **FRM Risk Scoping platform (Spring Boot, MySQL, React)** across **8 screens** and **30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70%** cut in manual reconciliation (**~ 2 weeks to $\sim 3\text{--}4$ days**).
- Designed an **11-table MySQL** schema on Spring Data JPA/Hibernate with clear controller / service / repository layers that auto-flags material items across **55 line items** and **14 entities**.
- Owned the Sheets to **MySQL** recon v2 migration across **18 files**, checking HFM extracts against public **10-Q** filings and fixing a join bug where two columns shared the same name.
- Led 3 engineers** through design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech** – Java, Spring Boot, Spring Data JPA, Hibernate, MySQL, JUnit, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built a menu pipeline (**Selenium $\rightarrow$ Kafka $\rightarrow$ Flink**) that brought onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month**.
- Built a **LangChain RAG + Gemini 2.5 Pro** extractor on **Milvus** that turns multilingual **PDF/image** menus into Uber Eats schema at **98% fidelity / 100% schema consistency** (offline eval).
- Raised successful ingestions to **95%+** with IP rotation, proxy pools, and retries when sites block scrapers.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Automated Uber driver and vehicle document checks for the **ANZ** region, achieving **99.9%** compliance with local authorities and eliminating about **20 hours/week** of manual verification.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Migrated a PHP (**Laravel**) monolith to **Spring Boot microservices** step by step for **1,500+ clients**, bringing p95 from **1.2s to 300ms** while mentoring **2 engineers**.
- Built a bulk e-invoice path on **Kafka** and **PostgreSQL** (tables split by tax quarter) that processed **1M+ government submissions/day** and **100K+/import**, lifting throughput from **700 to 4,000 requests/min**.
- Made IRP retries safe with **idempotency keys, retries, and a dead-letter queue**, and used Redis caching to lower repeat DB reads by **30%**.
- Set up **ELK and New Relic** alerting plus a client usage dashboard that lowered incident triage **70%** and support tickets **35%**, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Java, Spring Boot, Hibernate, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Spring Batch, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from PHP to **Spring Boot (Java)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on MySQL, MongoDB, Redis, and Elasticsearch.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** with scheduled jobs for video processing, reminders, and recording cleanup.

EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).